The Two Body Problem

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1 Formulae

1.1 Planetary Position

$$\tilde{\mathbf{r}} = (x, y) \tag{1}$$

1.2 Distance

$$|\mathbf{r}| = \sqrt{x^2 + y^2} \tag{2}$$

1.3 Two-Body Gravitational Acceleration

$$\tilde{\mathbf{a}} = -\frac{GM}{|r^3|}\vec{r} \tag{3}$$

Derived from Newton's Gravitational Attraction and the Law of Acceleration.
Here is Newton's Law of Gravitational Attraction:

$$\tilde{\mathbf{F}} = G\frac{m_1 m_2}{|\vec{r}|^3}\vec{r} \tag{4}$$

And here is the Second Law of Motion.

$$\tilde{\mathbf{F}} = m\vec{a} \tag{5}$$

If we combine these two, we will have an interesting derivation.
Assume m_1 is M.
Step 1:

$$m\vec{a} = G\frac{Mm}{|\vec{r}|^3}\vec{r} \tag{6}$$

Step 2:

$$\tilde{\mathbf{a}} = G\frac{Mm}{|\vec{r^3}|m}\vec{r} \tag{7}$$

The force $\vec{F}$ points to the radial vector, the gravity is attractive.
Hence why:

$$\tilde{\mathbf{a}} = -\frac{GM}{|r^3|}\vec{r} \tag{8}$$

1.4 Energy

$$\epsilon = \frac{v^2}{2} - \frac{\mu}{r} \tag{9}$$

However μ is switched to M due to the center mass being the object system is attracted to. Hence why:

$$\epsilon = \frac{v^2}{2} - \frac{M}{r} \tag{10}$$

1.5 Orbital Velocities

1.5.1 Circular

$$\mathbf{v_c} = \sqrt{\frac{GM}{r}} \tag{11}$$

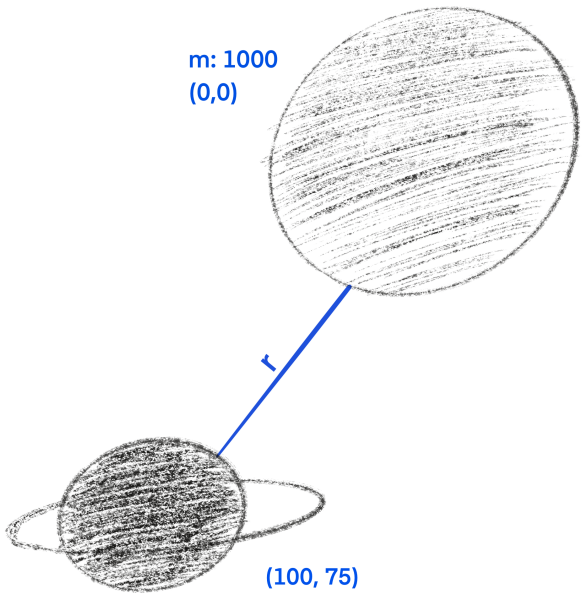
1.5.2 Escape

$$\mathbf{v_e} = \sqrt{\frac{2GM}{r}} \tag{12}$$

2 Problem Setup

Assume $G = 1$,

The Sun: $m = 1000$ at position $\vec{r} = (0, 0)$

The Planet: m is irrelevant, since it cancels out in the two-body derivation of Newtonian Gravitational Attraction. And the position is $\vec{r} = (100, 75)$.

The objects are placed at a distance r.